Final Project Codebook

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1 Data Dictionary

(Row) Data Label	Data Description
(1) `year`*	The year corresponding to a certain observation
(2) `line`**	The subway service corresponding to a certain observation
(3) `ind_weekend`	Indicator variable; takes the value 1 for weekends, and 0 for weekdays
(4) `yearly_total_ridership`*	The total ridership for the NYC subway in a year, in ten-thousands
(5) `yearly_adopt_budget`*	The adopted budget for the NYC Transit Authority in a year, in millions
(6) `count_roll_stock`	The number of distinct locomotive types serving a line
(7) `avg_roll_age`	The average locomotive age for a line, in years
(8) `range_roll_age`**	The difference in age between the newest and oldest locomotive, in years
(9) `avg_max_extent`	The average length from start-to-end of a subway service's routes, in miles
(10) `SWA_inc`	The station-weighted average of income for a line, in thousands of dollars
(11) `SWA_race`	The station-weighted average of non-white proportion for a line
(12) `ind_off_time`	Indicator variable; takes the value 0 for trains on a line that arrive at a station within five minutes of the scheduled arrival time

(Row)	Data type	Source file
(1)	Categorical	MTA Subway Terminal On-Time Performance: Beginning 2015
(2)	Categorical	MTA Subway Terminal On-Time Performance: Beginning 2015
(3)	Categorical binary	MTA Subway Terminal On-Time Performance: Beginning 2015
(4)	Continuous	MTA Daily Ridership Data: 2020 - 2025
(5)	Continuous	MTA Statement of Operations: Beginning 2019
(6)	Categorical	Wikipedia; this information can be derived from train service articles in the "rolling stock" section of the infobox
(7)	Continuous	Wikipedia; this information can be derived from train service articles in the "rolling stock" section of the infobox
(8)	Continuous	Wikipedia; this information can be derived from articles on specific trains used, found in the "rolling stock" section of the infobox
(9)	Continuous	MTA Subway Paths: Beginning 2019
(10)	Continuous	United States Census Bureau
(11)	Continuous	United States Census Bureau
(12)	Categorical binary	MTA Subway Terminal On-Time Performance: Beginning 2015

1.1 Appendix

Variables marked with an asterisk are not used in the final regression, and are dropped. Variables marked with a double asterisk are kept in the final data, but not used in regression.

Let i be the index for a subway line, t be the year, Boroughs be the set of boroughs, and X a statistic under consideration for a given borough b in year t (so, income and race). Then the station-weighted average SWA_{it} statistic for a line in a year is calculated as

$$SWA_{it} = \sum_{b \in \text{Boroughs}} \left(\frac{\#\text{ of stations } i \text{ serves in borough } b}{\text{total } \#\text{ of stations for } i} \times X_{bt} \right)$$

For example, suppose that the STAT train serves boroughs A and B with 2 stations in A and 3 in B, and that the average income of each borough in the year 2000 is 1 and 10 respectively. Then we have $SWA_{\text{STAT2000}} = \frac{2}{5} \times 1 + \frac{3}{5} \times 10 = \frac{32}{5}$.